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LEVER REDUCER

Abstract

Embodiments of a lever reducer and methods of using a lever reducer are provided. In some embodiments, the lever reducer system includes a first tower, a second tower, and a lever reducer configured to couple to the first tower and the second tower. The lever reducer can include a fulcrum configured to be coupled to the first tower and a swivel configured to be coupled to the second tower. The lever reducer is configured to vertically lift the second tower. The method can include coupling a first tower to a first vertebra, as well as coupling a second tower to a second vertebra. The method can include coupling a lever reducer to the first tower and the second tower by coupling a fulcrum to the first tower and a swivel to the second tower. The method can include applying a force to the lever reducer to vertically lift the second tower.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/723,932, filed Apr. 19, 2022, which claims priority benefit to U.S. Provisional Patent Application No. 63/180,931, filed Apr. 28, 2021, the disclosures of each are incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] The present disclosure relates to surgical instruments and methods for moving one component into an adjacent position to another component, in some cases moving a vertebra with an implanted fastener into an adjacent position to another vertebra with an implanted fastener.

Description of the Related Art

[0003] The vertebral column comprises a series of alternating vertebrae and fibrous discs that provide axial support and movement to the upper portions of the body. The vertebral column typically comprises thirty-three vertebrae, with seven cervical (C1-C7), twelve thoracic (T1-T12), five lumbar (L1-L5), five fused sacral (S1-S5) and four fused coccygeal vertebrae. Each vertebra includes an anterior body with a posterior arch. The posterior arch comprises two pedicles and two laminae that join posteriorly to form a spinous process. Projecting from each side of the posterior arch is a transverse, superior and inferior articular process. The facets of the superior and inferior articular processes form facet joints with the articular processes of the adjacent vertebrae.

[0004] The typical cervical vertebrae differ from the other vertebrae with relatively larger spinal canals, oval shaped vertebral bodies, bifid spinous processes and foramina in their transverse processes. These foramina transversaria contain the vertebral artery and vein. The first and second cervical vertebrae are also further differentiated from the other vertebrae. The first cervical vertebra lacks a vertebral body and instead contains an anterior tubercle. Its superior articular facets articulate with the occipital condyles of the skull and are oriented in a roughly parasagittal plane. The cranium is able to slide forward and backwards on this vertebra. The second cervical vertebra contains an odontoid process, or dens, which projects superiorly from its body. It articulates with the anterior tubercle of the atlas, forming a pivot joint. Side to side movements of the head occur at this joint. The seventh cervical vertebra is sometimes considered atypical since it lacks a bifid spinous process.

[0005] The typical lumbar vertebrae are distinguishable from the other vertebrae by the absence of foramina transversaria and the absence of facets on the surface of the vertebral body. The lumbar vertebral bodies are larger than the thoracic vertebral bodies and have thicker pedicles and laminae projecting posteriorly. The vertebral foramen is triangular in shape and larger than the foramina in the thoracic spine but smaller than the foramina in the cervical spine. The superior and inferior articular processes project superiorly and inferiorly from the pedicles, respectively. The sacrum is a large bone at the base of the spine formed by the fusion of the five sacral vertebrae (S1-S5).

[0006] In orthopedic surgery, and particularly in spinal surgery, it is well known to correct an injury, malformation, or other defect using an implanted rod affixed to a body part to be corrected. For example, rod systems have been developed for correcting the positioning of and stabilizing of the spine, and for facilitating fusion at various levels of the spine. In one such system, the rod or elongated implant can be disposed longitudinally along a length of the spine. The rod can be bent, either prior to or during surgery, to correspond to the normal curvature of the spine in the particular

region being instrumented, or to such other curvature as the surgeon may deem appropriate to correct the defect. For example, the rod can be bent to form a normal kyphotic curvature for the thoracic region of the spine, or to form a normal lordotic curvature for the lumbar region. The rod can then be attached or engaged to a number of fasteners which have been inserted or implanted into the vertebrae along the segment of the spinal column.

[0007] Fasteners are well known in the art and can include all types of bone screws, hooks, bolts, etc. configured to engage the vertebrae. For instance, one such fastener is a laminar hook, configured to engage a lamina of the vertebra. Another prevalent fastener is a spinal screw which can be threaded into a pedicle or other portion of vertebral bone. Examples of spinal screws include monoaxial spinal screws and polyaxial spinal screws.

[0008] In some spinal procedures, rods are coupled to two or more fasteners that are fixed to vertebrae, for instance at opposite sides of the spine or spinous processes. The fasteners can be threaded into a portion of several vertebral bodies, such as the pedicles of these vertebrae. The rod can be coupled to the bone screws to provide corrective and stabilizing forces to the spine. Affixing a rod to a fastener generally requires the rod to be in an adjacent position or in contact with the fastener. This may require that the rod and implanted fastener be moved with respect to each other so that the rod occupies space within a channel or other opening in the fastener. The rod can be coupled to the implanted fastener using a set screw, plug or other appropriate closure device. The process of placing a rod within or adjacent to an implanted fastener so that they can be coupled together is termed “reducing” the rod.

SUMMARY

[0009] In some embodiments, a lever reducer system is provided. The lever reducer system can include a first tower. The lever reducer system can include a second tower. The lever reducer system can include a lever reducer configured to couple to the first tower and the second tower. In some embodiments, the lever reducer comprises a fulcrum configured to be coupled to the first tower. In some embodiments, the lever reducer comprises a swivel configured to be coupled to the second tower. In some embodiments, the lever reducer is configured to vertically lift the second tower.

[0010] In some embodiments, the fulcrum is aligned with the first tower. In some embodiments, the fulcrum is configured to be at least partially inserted into the first tower. In some embodiments, the fulcrum is configured to be at least partially inserted into a lumen of the first tower through which a threaded cap was inserted. In some embodiments, the fulcrum is configured to translate relative to a base of the lever reducer. In some embodiments, the fulcrum is configured to rotate relative to a base of the lever reducer. In some embodiments, the fulcrum comprises a slidable pivot. In some embodiments, the fulcrum is configured to translate in a channel within a base of the lever reducer to accommodate different spans between the first tower and the second tower. In some embodiments, the fulcrum is configured to translate into one or more locking grooves that provide discrete translational positions. In some embodiments, the position of the swivel relative to the fulcrum is configured to allow for mostly vertical motion of the second tower. In some embodiments, the swivel is configured to rotate relative to a base of the lever reducer. In some embodiments, the swivel is configured to rotate in one direction, but rotation in the opposite direction is limited or prevented. In some embodiments, the lever reducer system can include a ratchet mechanism that maintains the vertical lift of the second tower. In some embodiments, a lumen of the second tower is accessible during vertical lift of the second tower.

[0011] In some embodiments, a method of using a lever reducer system is provided. The system can include coupling a first tower to a first vertebra. The system can include coupling a second tower to a second vertebra. The system can include coupling a lever reducer to the first tower and the second tower by coupling a fulcrum to the first tower and a swivel to the second tower. The system can include applying a force to the lever reducer to vertically lift the second tower.

[0012] In some embodiments, the first vertebra and the second vertebra are lumbar vertebrae. In

some embodiments, the method can include translating the fulcrum within a channel of a base of the lever reducer to accommodate different spans between the first tower and the second tower. In some embodiments, the method can include translating the fulcrum within a channel of a base of the lever reducer to accommodate different offsets between the first vertebra and the second vertebra. In some embodiments, the method can include translating the fulcrum into one or more locking grooves that provide discrete translational positions. In some embodiments, the method can include engaging a pawl with a gear to maintain the vertical lift of the second tower. In some embodiments, the method can include securing a rod to a fastener while the lever reducer is coupled to the first tower and the second tower. In some embodiments, the method can include securing a rod to a fastener before coupling the lever reducer to the first tower and the second tower.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] These and other features, aspects and advantages of the described embodiments are described with reference to drawings of certain preferred embodiments, which are intended to illustrate, but not to limit. It is to be understood that the attached drawings are for the purpose of illustrating concepts of the described embodiments and may not be to scale.

[0014] FIG. 1 illustrates a perspective view of a lever reducer system according to an embodiment.

[0015] FIG. 2 illustrates a perspective view of a lever reducer of FIG. 1.

[0016] FIG. 3 illustrates a top view of the lever reducer of FIG. 1.

[0017] FIG. 4 illustrates a bottom view of the lever reducer of FIG. 1.

[0018] FIG. 5 illustrates a front view of the lever reducer of FIG. 1.

[0019] FIG. 6 illustrates a first side view of the lever reducer of FIG. 1.

[0020] FIG. 7 illustrates a cross-sectional view of the lever reducer of FIG. 1.

[0021] FIG. 8 illustrates a fulcrum of the lever reducer of FIG. 1.

[0022] FIG. 9 illustrates a second side view of the lever reducer of FIG. 1.

[0023] FIG. 10 illustrates a cross-sectional view of the lever reducer of FIG. 1.

[0024] FIG. 11 illustrates an actuator of the lever reducer of FIG. 1.

[0025] FIG. 12 illustrates a first tower of the lever reducer system of FIG. 1.

[0026] FIG. 13 illustrates exploded components of the lever reducer system of FIG. 1.

[0027] FIG. 14 illustrates the first tower and a second tower of the lever reducer system of FIG. 1.

[0028] FIGS. 15A-15F illustrate movement of the lever reducer system of FIG. 1.

DETAILED DESCRIPTION

[0029] FIG. 1 illustrates a perspective view of an embodiment of a lever reducer system **100**. The lever reducer system **100** can include a lever reducer **102**. The lever reducer **102** can be utilized to move one component into an adjacent position to another component as described herein. The lever reducer **102** can be reversibly coupled to other components of the lever reducer system **100**. The lever reducer **102** can provide leverage to a user to vertically lift a vertebra. The lever reducer **102** can facilitate rod reduction by moving a fastener into alignment with a rod. The lever reducer **102** can have many uses and advantages, as described herein.

[0030] The lever reducer system **100** can include a first tower **104**. The first tower **104** can be any elongate structure. The first tower **104** can be a component of a tower based screw system. The first tower **104** can be a component of an extended tab screw system. The first tower **104** can be a component of a tower-based screw system or the extended tabs on minimally invasive pedicle screws with breakoff tabs. The first tower **104** can provide access through the skin and tissue of the patient toward a first vertebra. The first tower **104** can be coupled to the lever reducer **102**.

[0031] The lever reducer system **100** can include a first fastener **106**. The first fastener **106** can be

coupled to any portion of the spine. The first fastener **106** can be coupled to the first vertebra. The first fastener **106** can have any form or function to engage the vertebra. The first fastener **106** can couple to a pedicle. The first fastener **106** can be a polyaxial screw. The first tower **104** can couple to the first fastener **106**. The first tower **104** can couple to a head of the first fastener **106**. In some embodiments, the first tower **104** is a separate component from the first fastener **106**. The first tower **104** can easily couple and decouple from the first fastener **106**. The first tower **104** can include a locking feature to ensure attachment to the first fastener **106**. In some embodiments, the first tower **104** is a breakaway tower. In some embodiments, the first tower **104** is integrated with the corresponding first fastener **106**. The first tower **104** and the head of the first fastener **106** can be unitarily formed. The first tower **104** and the head of the first fastener **106** can include a score line therebetween. The first tower **104** can easily and cleanly break away at the score line, leaving the head of the first fastener **106** anchored to the first vertebra. The first tower **104** can include extended tabs from the head of the first fastener **106**. The extended tabs can be removed in a single motion by applying a force to the extended tabs. The first fastener **106** can be a minimally invasive pedicle screw. The first tower **104** can be breakoff extended tabs. The first tower **104** can provide access to the first fastener **106**.

[0032] The lever reducer system **100** can include a second tower **108**. The first tower **104** and the second tower **108** can be identical. The second tower **108** can be any elongate structure. The second tower **108** can be a component of a tower based screw system. The second tower **108** can be a component of an extended tab screw system. The second tower **108** can be a component of a tower-based screw system or the extended tabs on minimally invasive pedicle screws with breakoff tabs. The second tower **108** can provide access through the skin and tissue of the patient toward a second vertebra. The second tower **108** can be coupled to the lever reducer **102**. The second tower **108** can be lifted by the lever reducer **102**, as described herein.

[0033] The lever reducer system **100** can include a second fastener **110**. The first fastener **106** and the second fastener **110** can be identical. The second fastener **110** can be coupled to any portion of the spine. The second fastener **110** can be coupled to the second vertebra. The second fastener **110** can have any form or function to engage the vertebra. The second fastener **110** can couple to a pedicle. The second fastener **110** can be a polyaxial screw. The second tower **108** can couple to the second fastener **110**. The second tower **108** can couple to a head of the second fastener **110**. In some embodiments, the second tower **108** is a separate component from the second fastener **110**. The second tower **108** can easily couple and decouple from the second fastener **110**. The second tower **108** can include a locking feature to ensure attachment to the second fastener **110**. In some embodiments, the second tower **108** is a breakaway tower. In some embodiments, the second tower **108** is integrated with the corresponding second fastener **110**. The second tower **108** and the head of the second fastener **110** can be unitarily formed. The second tower **108** and the head of the second fastener **110** can include a score line therebetween. The second tower **108** can easily and cleanly break away at the score line, leaving the head of the second fastener **110** anchored to the second vertebra. The second tower **108** can include extended tabs from the head of the second fastener **110**. The extended tabs can be removed in a single motion by applying a force to the extended tabs. The second fastener **110** can be a minimally invasive pedicle screw. The second tower **108** can be breakoff extended tabs. The second tower **108** can provide access to the second fastener **110**.

[0034] The lever reducer system **100** can include a rod **112**. The rod **112** can be curved. The rod **112** can laterally extend through the first tower **104**. The rod **112** can laterally extend through the second tower **108**. The rod **112** can slide relative to the towers **104**, **108** and toward the vertebrae. The rod **112** can be seated within the head of the first fastener **106**. The rod **112** can be secured by a cap to the first fastener **106**. The rod **112** can be located within the second tower **108** when the rod **112** is secured to the first fastener **106**. The rod **112** can be vertically above the head of the second fastener **110**. The rod **112** cannot be secured to the second fastener **110** due to the position of the

second fastener **110**. The rod **112** cannot be secured to the second fastener **110** due to the position of the second vertebra.

[0035] The lever reducer **102** can lift the second tower **108**. The lever reducer **102** can lift the second fastener **110**. The lever reducer **102** can lift the second vertebra. The lever reducer **102** can raise the second fastener **110** such that the rod **112** can be seated within the head of the second fastener **110**.

[0036] The lever reducer **102** can raise the second vertebra toward a natural curvature. The lever reducer **102** can raise the second vertebra to correct spondylolisthesis or slippage of the second vertebra. The lever reducer **102** can raise the second vertebra to reduce compression of the spinal nerves. The lever reducer **102** can raise the second vertebra into alignment with the first vertebra.

[0037] In some methods, the first vertebra and the second vertebra can be any adjacent vertebrae. In some methods, the first vertebra and the second vertebra are not adjacent vertebrae and have one or more intermediate vertebra therebetween. The first vertebra and the second vertebra can be lumbar vertebrae. In some methods, the first vertebra and the second vertebra can be the L5 and L4 vertebra, respectively. In some methods, the first vertebra and the second vertebra can be the L4 and L5 vertebra, respectively. In some methods, the first vertebra is superior to the second vertebra. In some methods, the first vertebra is inferior to the second vertebra. In some methods, the first vertebra and the second vertebra can be the S1 and L5 vertebra, respectively. In some methods, the first tower **104** is coupled to the sacrum. In some methods, the first fastener **106** is coupled to the sacrum. In some methods, the second tower **108** is coupled to L5. In some methods, the second fastener **110** is coupled to L5. The lumbar vertebra L5 can be lifted relative to the sacrum. In some methods, the first vertebra and the second vertebra can be the L5 and S1 vertebra, respectively. In some methods, the first tower **104** is coupled to L5. In some methods, the first fastener **106** is coupled to L5. In some methods, the second tower **108** is coupled to the sacrum. In some methods, the second fastener **110** is coupled to the sacrum. The sacrum can be lifted relative to the lumbar vertebra L5. The first tower **104** and the first fastener **106** can be coupled to any portion of the spine. The second tower **108** and the second fastener **110** can be coupled to any portion of the spine.

[0038] The lever reducer system **100** described herein can be located at any level of the vertebral column. The fasteners **106**, **110** can be secured to any vertebrae. The rod **112** can be positioned between any vertebrae. In the description herein, the rod **112** is described as positioned between the first vertebra and the second vertebra. It should be appreciated that the lever reducer system **100** can be utilized in any portion of the spine.

[0039] The desired orientation of the rod **112** can depend on the first vertebra and the second vertebra. The rod **112** can restore a natural curvature of the spine. The rod **112** can provide a correction in one or more planes. The rod **112** can fuse the first vertebra and the second vertebra. The rod **112** can be placed at any angle to the transverse plane, including parallel, substantially parallel, perpendicular, substantially perpendicular, 0 degrees, 15 degrees, 30 degrees, 45 degrees, 60 degrees, 75 degrees, 90 degrees, or any range of two of the foregoing values. The rod can be placed at any angle to the sagittal plane, including parallel, substantially parallel, perpendicular, substantially perpendicular, 0 degrees, 15 degrees, 30 degrees, 45 degrees, 60 degrees, 75 degrees, 90 degrees, or any range of two of the foregoing values. The rod can be placed at any angle to the coronal plane, including parallel, substantially parallel, perpendicular, substantially perpendicular, 0 degrees, 15 degrees, 30 degrees, 45 degrees, 60 degrees, 75 degrees, 90 degrees, or any range of two of the foregoing values.

[0040] The lever reducer system **100** can facilitate placement of the rod **112** relative to the fasteners **106**, **110** by moving one component into an adjacent position to another component. The lever reducer system **100** can facilitate placement of the rod **112** relative to the fasteners **106**, **110** by moving the second vertebra with the second fastener **110** into an adjacent position relative to the first vertebra with the first fastener **106**. The lever reducer system **100** can facilitate placement of

the rod **112** by raising the second fastener **110** to be able to seat the rod **112** within the head of the second fastener **110**.

[0041] FIG. **2** illustrates a perspective view of the lever reducer **102**. FIG. **3** illustrates a top view of the lever reducer **102**. FIG. **4** illustrates a bottom view of the lever reducer **102**. FIG. **5** illustrates a front view of the lever reducer **102**.

[0042] The lever reducer **102** can include a handle **120**. The handle **120** can include a grip **122**. The grip **122** can be a knurled surface. The grip **122** can be a textured surface. The grip **122** can have any feature to facilitate manipulation by a user. The handle **120** can be removable. In some embodiments, the handle **120** can include a threaded tip. The handle **120** can couple to a base **124**. The base **124** can include a threaded bore to engage the threaded tip of the handle **120**. In some embodiments, the base **124** includes a threaded tip and the handle **120** includes a threaded bore. In some embodiments, the handle **120** and the base **124** are integrally or monolithically formed. The handle **120** can be angled relative to the base **124**. The handle **120** can form an angle relative to the base **124** of 0 degrees, 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, between 0 degrees and 20 degrees, between 10 degrees and 30 degrees, less than 45 degrees, less than 60 degrees, or any range of two of the foregoing values.

[0043] The lever reducer **102** can include a fulcrum **130**. The lever reducer **102** can include a first engagement feature **132**. The base **124** can include the first engagement feature **132**. The first engagement feature **132** can include a main translational channel **160**. The first engagement feature **132** can allow the fulcrum **130** to translate relative to the base **124**. The first engagement feature **132** can allow the fulcrum **130** to translate linearly. The first engagement feature **132** can allow the fulcrum **130** to translate a distance relative to the base **124**. In some embodiments, the fulcrum **130** can translate relative to the base **124** a distance of 0 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, 17 mm, 18 mm, 19 mm, 20 mm, 21 mm, 22 mm, 23 mm, 24 mm, 25 mm, 26 mm, 27 mm, 28 mm, 29 mm, 30 mm, between 5 mm and 15 mm, between 10 mm and 20 mm, between 15 mm and 30 mm, up to 10 mm, up to 15 mm, or any range of two of the foregoing values.

[0044] The first engagement feature **132** can allow the fulcrum **130** to rotate relative to the base **124**. The first engagement feature **132** can allow the fulcrum **130** to rotate and translate simultaneously. The first engagement feature **132** can allow the fulcrum **130** to rotate without translation. In some embodiments, the fulcrum **130** can rotate relative to the base **124** to form an angle of 0 degrees, 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 45 degrees and 90 degrees, between 30 degrees and 120 degrees, greater than 45 degrees, greater than 60 degrees, or any range of two of the foregoing values.

[0045] In some embodiments, the fulcrum **130** can rotate relative to the base **124** a greater degree in a distal position of the fulcrum **130**. The base **124** can include a recess **134**. The recess **134** can allow the fulcrum **130** to rotate relative to the base **124** to form an angle of 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, 130 degrees, 135 degrees, 140 degrees, 145 degrees, 150 degrees, 155 degrees, 160 degrees, 165 degrees, 170 degrees, 175 degrees, 180 degrees, 185 degrees, 190 degrees, 195 degrees, 200 degrees, 205 degrees, 210 degrees, 215 degrees, 220 degrees, 230 degrees, 235 degrees, 240 degrees, 245 degrees, 250 degrees, 255 degrees, 260 degrees, 265 degrees, 270 degrees, between 180 degrees and 270 degrees, between 90 degrees and 180 degrees, greater than 90 degrees, greater than 120 degrees, or any range of two of the foregoing values.

[0046] The fulcrum **130** can include a bearing surface **136**. The bearing surface **136** can rest against

the first tower **104**. The bearing surface **136** can include one or more ridges. The bearing surface **136** can be continuous around the fulcrum **130**. The bearing surface **136** can be a ring. The bearing surface **136** can be discontinuous around the fulcrum **130**. The bearing surface **136** can be a pair of tabs. The bearing surface **136** can be two diametrically opposed tabs.

[0047] The fulcrum **130** can include an insertion surface **138**. The insertion surface **138** can be below the bearing surface **136**. The insertion surface **138** can be inserted into the first tower **104**. The distal end of the fulcrum **130** can be tapered to facilitate insertion into the first tower **104**. The insertion surface **138** can have a tapered end. The insertion surface **138** can be cylindrical. The insertion surface **138** can be the same or similar cross-section as a cross-section of a lumen of the first tower **104**. The insertion surface **138** can have circular or rounded cross-section. In some embodiments, the fulcrum **130** can rotate within the first tower **104**. The fulcrum **130** can have an infinite number of rotational orientations relative to the first tower **104**. In some embodiments, the insertion surface **138** can have non-circular cross-section. The insertion surface **138** can have a keyed cross-section. In some embodiments, the fulcrum **130** cannot rotate within the first tower **104**. The fulcrum **130** can have a discrete number of rotational orientations relative to the first tower **104**. The fulcrum **130** can have one rotational orientation relative to the first tower **104**.

[0048] The lever reducer **102** can include a swivel **140**. The lever reducer **102** can include a second engagement feature **142**. The second engagement feature **142** can be ratchet mechanism. The second engagement feature **142** can include an actuator **144**. The second engagement feature **142** can include a pin **146**. The actuator **144** and the pin **146** can couple to the base **124**. The second engagement feature **142** can allow the swivel **140** to rotate relative to the base **124**. The second engagement feature **142** can prevent the swivel **140** from translating. The second engagement feature **142** can maintain the rotational position of the swivel **140**. The second engagement feature **142** can allow rotational motion of the swivel **140** in a first direction. The second engagement feature **142** can prevent rotational motion of the swivel **140** in a second direction, opposite the first direction. In some embodiments, the second engagement feature **142** can allow clockwise rotation of the swivel **140** but prevent counterclockwise rotation of the swivel **140**. The second engagement feature **142** can maintain a rotational position of the swivel **140**.

[0049] The second engagement feature **142** can allow the swivel **140** to rotate without translation. In some embodiments, the swivel **140** can rotate relative to the base **124** to form an angle of 0 degrees, 5 degrees, 10 degrees, 15 degrees, 20 degrees, 25 degrees, 30 degrees, 35 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees, 65 degrees, 70 degrees, 75 degrees, 80 degrees, 85 degrees, 90 degrees, 95 degrees, 100 degrees, 105 degrees, 110 degrees, 115 degrees, 120 degrees, between 45 degrees and 90 degrees, between 30 degrees and 120 degrees, greater than 45 degrees, greater than 60 degrees, or any range of two of the foregoing values. The second engagement feature **142** can allow the swivel **140** to travel in an arc.

[0050] The swivel **140** can include a gripping surface **148**. The gripping surface **148** can be inserted around the second tower **108**. The second tower **108** can include a keyed surface. The second tower **108** can have a polygonal surface including two flats, three flats, four flats, five flats, six flats, seven flats, eight flats, nine flats, ten flats, eleven flats, twelve flats, between six flats and eight flats, or any range of two of the foregoing values. The gripping surface **148** can have a keyed socket. The gripping surface **148** can have a polygonal surface including two flats, three flats, four flats, five flats, six flats, seven flats, eight flats, between two flats and four flats, or any range of two of the foregoing values. The gripping surface **148** can have the form of an open-end wrench.

[0051] The gripping surface **148** can be the same or similar cross-section as a cross-section of the second tower **108**. The gripping surface **148** can have non-circular cross-section. The gripping surface **148** can have a keyed cross-section. In some embodiments, the gripping surface **148** cannot rotate relative to the second tower **108**. The gripping surface **148** can have a discrete number of rotational orientations relative to the second tower **108**. The gripping surface **148** can have one rotational orientation relative to the second tower **108**. In some embodiments, the gripping surface

148 can have circular or rounded cross-section. The gripping surface **148** can rotate relative to the second tower **108**. The gripping surface **148** can have an infinite number of rotational orientations relative to the second tower **108**.

[0052] The swivel **140** can have an L-shape geometry. The swivel **140** can have an extension **150**. The extension **150** can align with the actuator **144**. The gripping surface **148** can be laterally offset. The gripping surface **148** can be offset from the actuator **144**. The gripping surface **148** can be perpendicular to the extension **150**.

[0053] The lever reducer **102** can have features aligned generally along a longitudinal axis of the lever reducer **102** that facilitate the application of a force. In some embodiments, the handle **120** can be aligned with the first engagement feature **132**. In some embodiments, the handle **120** can be aligned with the fulcrum **130**. This alignment can prevent a torque on the fulcrum **130** when a force is applied to the handle **120**. In some embodiments, the handle **120** can be aligned with the first tower **104**. This alignment can prevent a torque on the first tower **104** when a force is applied to the handle **120**. In some embodiments, the handle **120** can be aligned with the second tower **108**. This alignment can prevent a torque on the second tower **108** when a force is applied to the handle **120**. The first tower **104** and the second tower **108** can be aligned with each other. The shape of the swivel **140** can position the second tower **108** to be aligned with the first tower **104**.

[0054] The first tower **104** and the second tower **108** can be aligned with the fulcrum **130**. The fulcrum **130** is the support about which the lever reducer **102** pivots, as described herein. The handle **120** can be aligned with the fulcrum **130**. The handle **120** and the base **124** act as a lever to pivot about the fulcrum **130**. The handle **120** and the base **124** are a rigid body which pivots. As the handle **120** and a portion of the base **124** proximal to the fulcrum **130** are lowered, the portion of the base **124** distal to the fulcrum **130** raises.

[0055] The lever reducer **102** can include the first engagement feature **132** and the second engagement feature **142** which facilitate this pivoting. The first engagement feature **132** can allow rotation and translation of the fulcrum **130**. In some embodiments, the fulcrum **130** is a movable pivot. The second engagement feature **142** can allow rotation of the swivel **140**. In some embodiments, the swivel **140** allows rotation in a first direction but prevents or limits rotation in a second direction, opposite the first direction. Additional features of the first engagement feature **132** and the second engagement feature **142** are described herein.

[0056] FIG. 6 illustrates a first side view of the lever reducer **102**. FIG. 7 illustrates a cross-sectional view of the lever reducer **102**. FIG. 8 illustrates the fulcrum **130** of the lever reducer **102**.

[0057] The first engagement feature **132** can include the main translational channel **160**. The main translational channel **160** can be linear. The main translational channel **160** can be non-linear. The main translational channel **160** can form an arc. The first engagement feature **132** can include a pair of side channels **162**. The side channels **162** can extend along sides of the main translational channel **160**. The side channels **162** can extend the entire length of the main translational channel **160**. The side channels **162** can extend along a portion of the length of the main translational channel **160**. The first engagement feature **132** can include a pair of flanges **164**. The pair of flanges **164** can extend along sides of the main translational channel **160**. The pair of flanges **164** can extend under the side channels **162**. The pair of flanges **164** can extend the entire length of the main translational channel **160**. The pair of flanges **164** can extend along a portion of the length of the main translational channel **160**.

[0058] The fulcrum **130** can include a pair of pivots **166**. The pair of pivots **166** can be axially aligned. The pair of pivots **166** can be diametrically opposed. The pair of pivots **166** can form a pivot pin. The fulcrum **130** and the pair of pivots **166** can be integrally or monolithically formed. In some embodiments, the fulcrum **130** can include a bore configured to receive a pivot pin. The fulcrum **130** and the pair of pivots **166** or other pivot pin can be separately formed. The pair of pivots **166** can be cylindrical. The pair of pivots **166** can include rounded edges. The pair of pivots **166** can be semi-spherical.

[0059] The pair of pivots **166** extend into the pair of side channels **162** of the base **124**. Each pivot **166** slides along the side channels **162** as the fulcrum **130** slides. Each pivot **166** is retained within the side channel **162** by the flange **164** as the fulcrum **130** slides. Each pivot **166** slides along the flange **164**. The bearing surface **136** and the insertion surface **138** extend from the main translational channel **160** as the fulcrum **130** slides. The fulcrum **130** freely translates proximally and distally as the pair of pivots **166** is retained in the side channels **162**.

[0060] The pair of pivots **166** can rotate within the pair of side channels **162**. Each pivot **166** is retained within the side channel **162** by the flange **164** as the fulcrum **130** rotates. Each pivot **166** rotates relative to the flange **164**. The shape of the pair of pivots **166** can facilitate rotation. The fulcrum **130** can rotate relative to the base **124**. The fulcrum **130** can translate relative to the base **124**. The fulcrum **130** can rotate and translate relative to the base **124**.

[0061] In some embodiments, the pair of pivots **166** can translate within the first engagement feature **132** in discrete steps. The first engagement feature **132** can include one or more locking grooves that provide discrete translational positions. The first engagement feature **132** can include two locking grooves, three locking grooves, four locking grooves, five locking grooves, six locking grooves, seven locking grooves, eight locking grooves, nine locking grooves, ten locking grooves, or any range of two of the foregoing values. The first engagement feature **132** can include any number of locking grooves. The locking grooves can be formed in the pair of flanges **164**. The locking grooves can be formed in the pair of side channels **162**. The locking grooves can be formed in the main translational channel **160**. The locking grooves can be separately formed from the components of the first engagement feature **132**. The locking grooves can interact with the pair of pivots **166** of the fulcrum **130**. The pair of pivots **166** can slide distally and proximally to engage the next locking groove. The locking groove can apply a frictional force to retain the pair of pivots **166**. The pair of pivots **166** can overcome the frictional force by a force applied by the user. The pair of pivots **166** can overcome the frictional force by movement of one component relative to another.

[0062] The fulcrum **130** can include the bearing surface **136** configured to rest against the first tower **104**. The bearing surface **136** can include a pair of tabs. The pair of pivots **166** can be above the bearing surface **136**. The pair of pivots **166** can be aligned with the pair of tabs of the bearing surface **136**. The pair of pivots **166** and the bearing surface **136** can prevent or limit a left-right rocking motion of the fulcrum **130**. The insertion surface **138** can prevent or limit a left-right rocking motion of the fulcrum **130**. The pair of pivots **166** and the pair of side channels **162** can limit the motion of the fulcrum **130** to proximal-distal translation within the main translational channel **160**. The pair of pivots **166** and the side channels **162** can limit the motion of the fulcrum **130** to rotation within the main translational channel **160**. The rotation can be swinging in the proximal-distal direction. The shape of the fulcrum **130** can limit or reduce any other motion between the base **124** and the fulcrum **130**.

[0063] The lever reducer **102** can include a retention pin **168**. The retention pin **168** can prevent disengagement between the fulcrum **130** and the base **124**. The retention pin **168** can function as a proximal stop for the translation of the fulcrum **130**. The retention pin **168** can be inserted after the pair of pivots **166** are inserted within the pair of side channels **162**.

[0064] FIG. **9** illustrates a second side view of the lever reducer **102**. FIG. **10** illustrates a cross-sectional view of the lever reducer **102**. FIG. **11** illustrates the actuator **144** of the lever reducer **102**.

[0065] The second engagement feature **142** can include the actuator **144**. The actuator **144** can be actuated by the user. The actuator **144** can be automatically actuated. The actuator **144** can include a lever arm **170**. The lever arm **170** can be biased. The actuator **144** can include a spring **172**. The actuator **144** can include a pivot pin **174**. The lever arm **170** can pivot relative to the pivot pin **174**. The pivot pin **174** can couple the lever arm **170** to the base **124**. As the lever arm **170** is pivoted, the spring **172** can be compressed. The lever arm **170** can include a pawl **176**. The pawl **176** can

function as a ratchet.

[0066] The swivel **140** can include one or more gears **178**. The gears **178** can be radial gears. The gears **178** can be ramped. The gears **178** can form a wedge. The gears **178** can be asymmetric. The gears **178** can allow movement in one direction but prevent movement in another direction. The gears **178** can allow the pawl **176** to slide along the ramped surface of the gears **178** and into the next gear. The gears **178** can prevent the pawl **176** from moving in the opposite direction. The ramped surface of the gears **178** forms a wall or wedge to prevent movement of the pawl **176** in the opposite direction.

[0067] The second engagement feature **142** can include the pin **146**. The pin **146** can couple the swivel **140** to the base **124**. The swivel **140** can rotate relative to the base **124**. The swivel **140** can rotate in a first direction. The pawl **176** slides along a gear **178** thereby depressing the spring **172**. The pawl **176** slides along a gear **178** until falling to the next gear **178**. The spring **172** biases the lever arm **170** down into engagement with the groove of the gear **178**. The pawl **176** slides along the next gear **178** thereby depressing the spring **172**. The pawl **176** slides along a gear **178** until falling to the next gear **178**. The pawl **176** freely advances in a first direction. The pawl **176** can be limited or prevented from advancing in the opposite direction. The shape of the gears **178** provide a wedge preventing rotational movement in the opposite direction. The pawl **176** cannot vertically clear the wedge of the gear **178**. The spring **172** biases the pawl **176** downward preventing rotational movement in the opposite direction.

[0068] The second engagement feature **142** can allow the swivel **140** to rotate relative to the base **124** in one direction. The second engagement feature **142** provides a ratchet that limits rotational movement. The shape of the pawl **176** and the gears **178** allows rotation in one direction, but limits rotation in the opposite direction. The biasing force of the spring **172** allows rotation in one direction, but limits rotation in the opposite direction. The swivel **140** rotates about the pin **146** relative to the base **124**. The swivel **140** can form a slight arc during vertical motion as described herein. The second engagement feature **142** can prevent the swivel **140** from translating relative to the base **124**.

[0069] The second engagement feature **142** can maintain the rotational position of the swivel **140**. The second engagement feature **142** can maintain the correction between the first tower **104** and the second tower **108**. The second engagement feature **142** can maintain the correction between the vertebrae as the second vertebra is lifted. The shape of the pawl **176** and the gears **178** can maintain the rotation position during vertical lift. The biasing force of the spring **172** can maintain the rotation position during vertical lift.

[0070] FIG. **12** illustrates the first tower **104** of the lever reducer system **100**. FIG. **13** illustrates exploded components of the lever reducer system **100**. FIG. **14** illustrates the first tower **104** and the second tower **108** of the lever reducer system **100**. The first tower **104** and the second tower **108** facilitate placement of the rod **112**. The rod **112** can be generally straight. The rod **112** can include a bend. The bend can correspond to a bend of the vertebrae. The bend can correspond to a desired curvature of the spine. The bend can correspond to a desired spinal correction. The bend can correspond to a desired lift of the second vertebra.

[0071] In some methods, the first fastener **106** is coupled to the first vertebra. The first vertebra can be any vertebra along the spine of the patient. The first fastener **106** can include a threaded shaft **180**. The first fastener **106** can be rotated to engage the first vertebra. The first fastener **106** can be positioned to engage cortical bone. In some methods, the first fastener **106** engages the pedicle. The pedicle is a portion of the vertebra between the transverse process and the vertebral body. The pedicle can be used as an anchoring point for fasteners in spinal procedures. In other methods, the first fastener **106** engages another portion of the first vertebra. The first fastener **106** can couple to the spinal process. The first fastener **106** can couple to the transverse process. The first fastener **106** can couple to the lamina. The first fastener **106** can be coupled to any portion of the first vertebra.

[0072] The first fastener **106** can include a head **182**. The head **182** can include a slot **184** sized and

configured to receive the rod **112**. The head **182** can have polyaxial movement relative to the threaded shaft **180**. The threaded shaft **180** can have a bulbous surface **186**. The head **182** can receive the bulbous surface **186** of the threaded shaft **180**. The head **182** can move within a range of motion relative to the threaded shaft **180**. The head **182** can rotate relative to the threaded shaft **180**. The head **182** can tilt relative to the threaded shaft **180**. The polyaxial movement between the threaded shaft **180** and the head **182** can facilitate a range of orientations for the first fastener **106**. The polyaxial movement between the threaded shaft **180** and the head **182** can facilitate a range of orientations for the rod **112**. In some embodiments, the head **182** can have no movement relative to the threaded shaft **180**. In some embodiments, the head **182** can have only monoaxial movement relative to the threaded shaft **180**.

[0073] The first tower **104** can couple to the first fastener **106**. The first tower **104** can couple to the first fastener **106** after the first fastener **106** is coupled to the first vertebra. In other methods, the first tower **104** can couple to the first fastener **106** before the first fastener **106** is coupled to the first vertebra. The first tower **104** can couple to the head **182**. The head **182** can include a pair of recesses **188**. The first tower **104** can include a pair of arms **190**. The pair of arms **190** can engage the pair of recesses **188** to couple the first tower **104** to the first fastener **106**. The pair of arms **190** can be deflected radially outward to accommodate the head **182** therebetween. The pair of arms **190** can be biased inward into engagement with the pair of recesses **188**.

[0074] In some embodiments, the first tower **104** and the head **182** are integrally or monolithically formed. In some embodiments, the first tower **104** and the first fastener **106** are integrally or monolithically formed. The threaded shaft **180** can have no movement relative to the combined head **182** and the tower **104**. The threaded shaft **180** can have monoaxial movement relative to the combined head **182** and the tower **104**. The threaded shaft **180** can have polyaxial movement relative to the combined head **182** and the tower **104**. The first tower **104** can decouple from the first fastener **106** after the rod **112** is secured. The first tower **104** can include a score line indicating that location of the decoupling.

[0075] The first tower **104** can include a shaft **192**. The shaft **192** can include a slot **194** sized to receive the rod **112**. In some methods, the rod **112** can slide toward the first fastener **106**. In some methods, the rod **112** is inserted into the head **182** without sliding vertically in the slot **194**. In some methods, the rod **112** is inserted along a trajectory.

[0076] The first tower **104** can include a tower cap **196**. The tower cap **196** can rotate relative to the shaft **192**. The tower cap **196** can translate relative to the shaft **192**. The tower cap **196** can rotate and translate to control the engagement of the pair of arms **190** with the pair of recesses **188**. The tower cap **196** can rotate and translate to control the coupling of the first tower **104** and the first fastener **106**. The tower cap **196** can rotate and translate to disengage the pair of arms **190** with the pair of recesses **188**. The tower cap **196** can rotate and translate to control the decoupling of the first tower **104** and the first fastener **106**.

[0077] In some methods, the rod **112** is seated within the first fastener **106**. The shaft **192** and the tower cap **196** can form a lumen therethrough. The lumen can accommodate a threaded cap **198**. The threaded cap **198** can be lowered toward the first fastener **106**. The threaded cap **198** can couple to the first fastener **106** to retain the rod **112**. The threaded cap **198** can couple the rod **112** to the first fastener **106**. The threaded cap **198** can exert a force on the rod **112** to seat the rod **112** within the slot **184** of the head **182**. The threaded cap **198** can exert a force on the rod **112** which can exert a force on the bulbous surface **186**. The threaded cap **198** can exert a force on the rod **112** to prevent or limit the polyaxial movement of the threaded shaft **180**.

[0078] In some methods, the second fastener **110** is coupled to the second vertebra. The second vertebra can be adjacent to the first vertebra. The second vertebra can be separated from the first vertebra by one or more intermediate vertebra. The second vertebra can be superior to the first vertebra. The second vertebra can be inferior to the first vertebra. The second vertebra can be any vertebra along the spine of the patient. The second fastener **110** can include any feature of the first

fastener **106** described herein. The second tower **108** can couple to the second fastener **110**. The second tower **108** can be integrally formed with the second fastener **110**. The second tower **108** can include any of the features of the first tower **104** described herein. The rod **112** can extend through the slot **194** sized to receive the rod **112** of the second tower **108**.

[0079] The rod **112** is vertically displaced relative to the head **182** of the second fastener **110**. The rod **112** is not seated within the head **182** of the second fastener **110**. The rod **112** can be vertically lifted a distance from the desired position within the head **182** of the second fastener **110**. The distance can be 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, 17 mm, 18 mm, 19 mm, 20 mm, 21 mm, 22 mm, 23 mm, 24 mm, 25 mm, 26 mm, 27 mm, 28 mm, 29 mm, 30 mm, between 5 mm and 20 mm, between 10 mm and 30 mm, at least 5 mm, at least 10 mm, or any range of two of the foregoing values. The rod **112** is within the slot **184** of the head **182** when properly seated. The rod **112** exerts a force on the bulbous surface **186** when properly seated. The rod **112** can be secured by the threaded cap **198** when properly seated. The lever reducer **102** can be utilized to change the position of the second tower **108**. The lever reducer **102** can be utilized to change the position of the second fastener **110**. The lever reducer **102** can be utilized to change the position of the second vertebra. In some methods, the lever reducer **102** does not change the position of the first tower **104**, the first fastener **106**, or the first vertebra. In some methods, the lever reducer **102** does not change the position of the rod **112**.

[0080] In some methods, the rod **112** is not seated within the first fastener **106** before use of the lever reducer **102**. The first fastener **106** can be coupled to the first vertebra. The second fastener **110** can be coupled to the second vertebra. If separate components, the first tower **104** can couple to the first fastener **106**. If separate components, the second tower **108** can couple to the second fastener **110**. The towers **104**, **108** can be components of a tower-based screw system. If integrated components, the first fastener **106** and the first tower **104** are coupled to the first vertebra. If integrated components, the second fastener **110** and the second tower **108** are coupled to the second vertebra. The towers **104**, **108** can be extended tabs. The towers **104**, **108** can be integrated tabs on the fasteners **108**, **110**. The fasteners **108**, **110** and the towers **104**, **108** can be minimally invasive pedicle screws with breakoff tabs. The rod **112** is not positioned relative to the first fastener **106** before use of the lever reducer **102**. The threaded cap **198** is not positioned relative to the first fastener **106** before use of the lever reducer **102**. The rod **112** is not retained relative to the first fastener **106**. The lever reducer **102** can be utilized without the rod **112** in position. The lever reducer **102** can lift the second tower **108** relative to the first tower **104**. The lever reducer **102** can be utilized prior to introduction of the rod **112**. The lever reducer **102** can be used to lift the second tower **108** to align the first tower **104** and the second tower **108** prior to introduction of the rod **112**. The lever reducer **102** can be used to lift the head **182** of the second fastener **110**. The head of the second fastener **110** and the head of the first fastener **106** can be aligned prior to introduction of the rod **112**. The lever reducer **102** can change the vertical displacement of the second fastener **110** to allow introduction of the rod **112** therethrough.

[0081] In some methods, the rod **112** can be lowered after use of the lever reducer **102**. The rod **112** can be lowered along the first tower **104**. The rod **112** can be lowered along the second tower **108**. The rod **112** can be lowered along the first tower **104** and the second tower **108** simultaneously. The rod **112** can be seated within the first fastener **106**. The rod **112** can be seated within the second fastener **110**. The rod **112** can be seated within the first fastener **106** and seated within the second fastener **110** simultaneously. The rod **112** can be seated within the first fastener **106** and seated within the second fastener **110** sequentially. The rod **112** can be seated within the fasteners **106**, **110** by lowering the rod **112** along the towers **104**, **108**. The rod **112** can be seated within the fasteners **106**, **110** without lowering the rod **112** along the towers **104**, **108**. The rod **112** can be introduced after the second tower **108** is vertically lifted.

[0082] The rod **112** can be secured to the second fastener **110** while the lever reducer **102** is in use.

The threaded cap **198** can secure the rod **112** to the second fastener **110**. The lever reducer **102** can be removed. In some methods, the rod **112** can be secured to the first fastener **106** after the lever reducer **102** is removed. The threaded cap **198** can secure the rod **112** to the first fastener **106**. [0083] FIGS. **15A-15F** illustrate movement of the lever reducer system **100**. FIG. **15A** illustrates an initial position of the lever reducer system **100** according to some methods. FIGS. **15B-15E** illustrate intermediate positions of the lever reducer system **100** as the handle **120** is lowered according to some methods. FIG. **15F** illustrates a final position of the lever reducer system **100** according to some methods.

[0084] The first fastener **106** is secured to the first vertebra. The second fastener **110** is secured to the second vertebra. The fasteners **106, 110** can be secured in any order. The fasteners **106, 110** can be secured in series or simultaneously. The fasteners **106, 110** can allow for polyaxial movement that accommodates varying anatomical situations. In some methods, the fasteners **106, 110** are coupled to the respective pedicles of the first vertebra and the second vertebra. The first tower **104** can be secured to the first fastener **106** if separate components. The second tower **108** can be secured to the second fastener **110** if separate components. The towers **104, 108** can be secured in any order. The towers **104, 108** can be secured in series or simultaneously. The towers **104, 108** can be secured to the fasteners **106, 110** before securing the fasteners to the vertebrae. The towers **104, 108** can be secured to the fasteners **106, 110** after securing the fasteners to the vertebrae. In some methods, the towers **104, 108** and the fasteners **106, 110** are integrally formed.

[0085] The rod **112** can be positioned relative to the slot **194** in the first tower **104**. The rod **112** can be positioned relative to the slot **194** in the second tower **108**. The rod **112** can be seated within the first fastener **106**. The rod **112** can extend to the second tower **108**. The threaded cap **198** can be lowered down the first tower **104** to the first fastener **106**. The rod **112** can be secured to the first fastener **106**. In some methods, the rod **112** is secured to the first fastener **106** before use of the lever reducer **102**. In some methods, the threaded cap **198** is secured to the first fastener **106** before use of the lever reducer **102**. The rod **112** is coupled to the first fastener **106**. The rod **112** extends through the second tower **108**. The rod **112** can be vertically offset relative to the second fastener **110** preventing coupling of the rod **112** to the second fastener **110**. The rod **112** cannot be secured to the second fastener **110** due to the position of the relative components.

[0086] In some methods, the first vertebra is the L5 vertebra. In some methods, the second vertebra is the L4 vertebra. In some methods, the first vertebra is the L4 vertebra. In some methods, the second vertebra is the L5 vertebra, The L5 vertebra is disposed between the L4 vertebra and the sacrum. In some methods, the first vertebra is the S1 vertebra. In some methods, the second vertebra is the L5 vertebra. In some methods, the first vertebra is the L5 vertebra. In some methods, the second vertebra is the S1 vertebra. In some methods, the first vertebra is any sacral vertebra. In some methods, the second vertebra is any other vertebra. In some methods, the first vertebra is any lumbar vertebra. In some methods, the second vertebra is any other lumbar vertebra or any other vertebra. In some methods, the first vertebra is any thoracic vertebra. In some methods, the second vertebra is any other thoracic vertebra or any other vertebra. In some methods, the first vertebra is any cervical vertebra. In some methods, the second vertebra is any other cervical vertebra or any other vertebra. The first vertebra can be any vertebra along the length of the spine. The second vertebra can be any other vertebra along the length of the spine. The first vertebra and the second vertebra can be adjacent. The first vertebra and the second vertebra can be separated by one or more intermediate vertebra.

[0087] The towers **104, 108** are coupled to the fasteners **106, 110**. Movement of the second tower **108** causes corresponding movement of the second fastener **110**. Movement of the second tower **108** causes corresponding movement of the second vertebra. In some methods, the rod **112** can be secured to the first fastener **106** throughout movement of the second tower **108** and the second fastener **110**. The rod **112** can be disposed within the second tower **108** throughout movement of the second tower **108** and the second fastener **110**. The second tower **108** can slide relative to the

rod **112** during movement of the second tower **108** and the second fastener **110**. The second fastener **110** can be lifted toward the rod **112** during movement of the second tower **108** and the second fastener **110**. In some methods, the rod **112** is not secured to the first fastener **106** during use of the lever reducer **102**. The rod **112** can be secured after the second tower **108** is lifted. The rod **112** can be positioned relative to the aligned heads **182** of the fasteners **106**, **108** after the second tower **108** is lifted.

[0088] The second vertebra can be misaligned relative to the spinal column. The second vertebra can be misaligned due to spondylolisthesis or slippage of the second vertebra. The second vertebra can be slipped relative to a natural curvature. The second vertebra can cause compression of the spinal nerves extending through the foramen. The second vertebra can cause pain due to nerve compression. The second vertebra can have a sagittal deformity. The second vertebra can have a coronal deformity. The second vertebra can have a transverse deformity. The second vertebra can be misaligned relative to the first vertebra due to a deformity or degenerative condition.

[0089] The lever reducer **102** can engage the first tower **104**. The fulcrum **130** can be at least partially inserted into a lumen of the first tower **104**. The fulcrum **130** can be inserted into the tower cap **196** of the first tower **104**. In some methods, before the fulcrum **130** is inserted, the threaded cap **198** is inserted into the lumen of the first tower **104**. The threaded cap **198** can be rotated to secure the rod **112** to the first fastener **106**. The fulcrum **130** can utilize the same lumen as the threaded cap **198**.

[0090] The lever reducer **102** can engage the second tower **108**. The gripping surface **148** can engage the second tower **108**. The gripping surface **148** can grip a keyed surface below the tower cap **196** of the second tower **108**. The pawl **176** can engage a gear **178**. The actuator **144** can be biased by the spring **172**. The pawls **176** can be seated in a groove between gears **178**. Movement in one direction is allowed by overcoming the biasing force of the spring **172**. Movement in the opposite direction is prevented due to the shape of the gears **178**. The pawl **176** can maintain a rotational orientation between the towers **104**, **108**.

[0091] The fulcrum **130** can slide within the first engagement feature **132**. The first engagement feature **132** can accommodate the relative spacing between the first tower **104** and the second tower **108**. The bearing surface **136** of the fulcrum **103** can rest against the tower cap **196**. The insertion surface **138** of the fulcrum can be inserted into the first tower **104**. The first engagement feature **132** can allow the fulcrum **130** to translate relative to the base **124**. The fulcrum **130** can include a pair of pivots **166**. The pair of pivots **166** are retained within the side channel **162** as the fulcrum **130** slides in the main translational channel **160**. The fulcrum **130** can be at any location along the main translational channel **160**. The fulcrum **130** can be at a distal location. The fulcrum **130** can be at a proximal location. The fulcrum **130** can slide to accommodate the spacing between the first tower **104** and the second tower **108**. The fulcrum **130** can slide to accommodate the spacing between the first vertebra and the second vertebra. The fulcrum **130** can slide to accommodate the anatomy of the patient. In some embodiments, the fulcrum **130** is a slidable pivot. In some embodiments, the fulcrum **130** is a fixed pivot.

[0092] The pair of pivots **166** define the pivot point **200** of the lever reducer **102**. The pivot point **200** is movable as the fulcrum **130** moves. The pivot point **200** can slide relative to the base **124**. The pivot point **200** can be at any position. The fulcrum **130** can slide based on the orientation of the first tower **104** coupled to the first fastener **106**. The polyaxial orientation of the first fastener **106** can be locked when the rod **112** is secured to the first fastener **106**. The threaded cap **198** can lock the orientation of the first fastener **106**. The first tower **104** can be locked relative to the first fastener **106**. The lever reducer **102** can accommodate the fixed relationship between the first vertebra, the first fastener **106**, and the first tower **104**. The fulcrum **130** can slide to accommodate the fixed orientation of the first tower **104** relative to the first vertebra. In some methods, the first tower **104** is fixed relative to the threaded shaft **180** of the first fastener **106** when the rod **112** is secured. In some methods, the pivot point **200** is movable to limit or prevent applying a force or

torque to the threaded shaft **180** of the first fastener **106**. The pivot point **200** can be aligned with the first vertebra. In some embodiments, the orientation between the pivot point **200**, the first tower **104**, the first fastener **106**, and the first vertebra can be fixed. In some methods, the base **124** of the lever reducer **102** can translate relative to this pivot point **200**. In some methods, the base **124** of the lever reducer **102** can rotate relative to this pivot point **200**.

[0093] The lever system **100** can include a radial distance **202**. The radial distance **202** can be defined between the pivot point **200** and the pin **146**. The pin **146** can couple the swivel **140** to the base **124**. In some methods, the radial distance **202** is constant during movement of the second tower **108**. In some methods, the radial distance **202** defines an arc of movement of the swivel **140**. In some methods, the radial distance **202** varies during movement of the second tower **108**.

[0094] The handle **120** and the base **124** act as a lever relative to the pivot point **200**. The user can grip the handle **120**. The user can lower the handle **120** as shown in FIGS. **15A-15B**. The user can apply a force to the handle **120**. The user can apply a downward force. The lever reducer **102** can amplify the input force of the user to a greater output force to lift the vertebra. The lever reducer **102** can provide leverage to facilitate the vertical lift. The lever reducer **102** can lift the swivel **140** coupled to the second vertebra. The swivel **140** can move in an arc relative to the pivot point **200**. The swivel **140** can move mostly vertically. The swivel **140** can move slightly outward relative to the first tower **104**. The user can continue to lower the handle **120** as shown in FIGS. **15C-15E**. The second tower **108** can gradually lift. The second tower **108** can lift relative to the first tower **104**.

[0095] FIG. **15F** illustrates a final position of the second tower **108** according to some methods. In some methods, the first tower **104** and the second tower **108** can have the same height. The second tower **108** can be vertically lifted. In some methods, the first tower **104** and the second tower **108** can have different heights. The second tower **108** can be vertically lifted less than the height of the first tower **104**. The second tower **108** can be vertically lifted greater than the height of the first tower **104**.

[0096] The second tower **108** is gripped by the lever reducer **102** to not obstruct the lumen of the second tower **108**. The design allows clearance for an inserter for the threaded cap **198** to be placed in the second tower **108**. The second tower **108** is the tower in which rod **112** is being reduced. The threaded cap **198** can be inserted into the second tower **108** and locked to the second fastener **110**. The threaded cap **198** can secure the rod **112** to the second fastener **110**. The threaded cap **198** can limit or prevent polyaxial movement of the second fastener **110**. The threaded cap **198** lock the correction before removing lever reducer **102**.

[0097] In some methods, a third fastener and a third tower are coupled to the first vertebra. The first fastener **106** and the first tower **104** can be on one side of the first vertebra. The third fastener and the third tower can be coupled to the other side of the first vertebra. In some methods, a fourth fastener and a fourth tower are coupled to the second vertebra. The second fastener **110** and the second tower **108** can be on one side of the second vertebra. The fourth fastener and the fourth tower can be coupled to the other side of the second vertebra. The third tower and the fourth tower can be coupled to the lever reducer **102**. In some methods, two lever reducers **102** are utilized to lift the second vertebra. The third fastener and the fourth fastener can have any features of the fasteners described herein. The third tower and the fourth tower can have any features of the towers described herein.

[0098] The fasteners **106**, **110** can be coupled to the rod **112** to maintain the correction. The lever reducer **102** can be decoupled from the towers **104**, **108**. The towers **104**, **108** can be decoupled from the fasteners **106**, **110**. In some embodiments, the fasteners **106**, **110** and the towers **104**, **108** are integrally formed. The towers **104**, **108** can be snapped off from the respective fasteners.

[0099] The lever reducer system **100** can include several advantages. The lever reducer **102** can easily couple to the towers **104**, **108**. The lever reducer **102** can include a fulcrum **130** that can be at least partially inserted into the top of the first tower **104**. The fulcrum **130** can include a bearing surface **136** that rests against the first tower **104**. The fulcrum **130** can include a portion that is

inserted into the first tower **104**. The fulcrum **130** defines the pivot point **200** of the lever reducer system **100**. The lever reducer **102** pivots relative to the fulcrum **130**. The lever reducer **102** pivots relative to the first tower **104**. In some methods, the first tower **104** is not raised or lowered. The first tower **104** remains stationary. The first fastener **106** remains stationary. The first vertebra remains stationary. The pivot point **200** is aligned with the first tower **104**, the first fastener **106**, and the first vertebra.

[0100] The fulcrum can include a pair of pivot pins **166** about which the handle **120** and the base **124** pivot. As the handle **120** is lowered, a distal portion of the base **124** is raised. The distal portion of the base **124** can include the swivel **140**. The swivel **140** can couple to the second tower **108**. The lever reducer **102** pivots relative to the fulcrum **130** to raise the second tower **108**. The second tower **108** moves with the distal portion of the base **124** of the lever reducer **102**.

[0101] The second tower **108** vertically lifts. The second fastener **110** vertically lifts. The second vertebra vertically lifts. The second tower **108** moves in a slight arc relative to the first tower **104**. The slight arc can be beneficial for moving the second vertebra relative to the first vertebra. The slight arc can move the second vertebra away from the first vertebra. The slight arc can move the second tower **108** away from the first tower **104**. The slight arc can facilitate the alignment of anatomical structures. In some methods, the slight arc can be beneficial over a lift that is only vertical. In some methods, the slight arc can facilitate movement of the second vertebra relative to the first vertebra to avoid or limit interaction between bony prominences on the first vertebra and/or the second vertebra.

[0102] The pin **146** of the swivel **140** can define a hinge. The hinge is positioned relative to the fulcrum **130** to allow for mostly vertical lift of a vertebra. The hinge and the fulcrum **130** can be separated by the radial distance **202**. The radial distance **202** can be maintained as the second tower **108** lifts. In some embodiments, the radial distance **202** can allow a ratio of lateral displacement to vertical lift. In some embodiments, the lateral displacement is small allowing for mostly vertically lift. In some embodiments, the movement of the swivel **140** along the arc can allow for 1 mm lateral displacement relative to 3 mm vertical lift, 5 mm vertical lift, 7 mm vertical lift, 10 mm vertical lift, 12 mm vertical lift, 15 mm vertical lift, 17 mm vertical lift, 20 mm vertical lift, or 1 mm lateral displacement relative to any range of two of the foregoing values. In some embodiments, the movement of the swivel **140** along the arc can allow for a lateral displacement from 1 mm to 5 mm and a vertical displacement from 3 mm to 30 mm. In some embodiments, the movement of the swivel **140** along the arc can allow for a lateral displacement of 0 mm, 0.5 mm, 1 mm, 1.5 mm, 2 mm, 2.5 mm, 3 mm, 3.5 mm, 4 mm, 4.5 mm, 5 mm, between 1 mm and 2 mm, between 1 mm and 3 mm, between 1 mm and 4 mm, between 1 mm and 5 mm, greater than 1 mm, greater than 2 mm, less than 5 mm, less than 4 mm, or any range of two of the foregoing values. In some embodiments, the movement of the swivel **140** along the arc can allow for a vertical displacement of 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, 17 mm, 18 mm, 19 mm, 20 mm, 21 mm, 22 mm, 23 mm, 24 mm, 25 mm, 26 mm, 27 mm, 28 mm, 29 mm, 30 mm, between 5 mm and 10 mm, between 10 mm and 15 mm, between 15 mm and 20 mm, between 20 mm and 25 mm, between 25 mm and 30 mm, between 1 mm and 10 mm, between 10 mm and 20 mm, between 20 mm and 30 mm, or any range of two of the foregoing values.

[0103] The fulcrum **130** can translate in the main translational channel **160** within the base **124**. The fulcrum **130** can translate to quickly accommodate different spans between towers **104**, **108**. The fulcrum **130** can translate to quickly accommodate different offsets between vertebral bodies. The fulcrum **130** can translate to allow the lever reducer **102** to be used for patients having different sized vertebra. The fulcrum **130** can translate to allow the lever reducer **102** to be used at any level of the spine. The fulcrum **130** can translate to allow the lever reducer **102** to be used to accommodate a variety of degenerative conditions of the vertebrae.

[0104] The second engagement feature **142** maintains correction during the reduction maneuver.

The second engagement feature **142** can include a ratchet mechanism with the pawl **176**. The pawl **176** engages gears **178**. The pawl **176** can be biased in between gears **178** in an initial position of the towers **104**, **108**. The pawl **176** can allow movement in only one direction. The gears **178** can allow movement in only one direction. The pawl **176** can allow movement of the second tower **108** away from the first tower **104**. The pawl **176** can allow the swivel **140** to follow the arc defined by the radial distance **202**. The pawl **176** can prevent movement in another direction based on the shape of the gears **178**. The pawl **176** can prevent movement of the second tower **108** in a direction opposite to the direction of vertical lift. The pawl **176** can maintain a correction during vertical lift. The pawl **176** can maintain the vertical lift of the second tower **108**. The ratchet mechanism can maintain the vertical lift of the second tower **108**.

[0105] The pawl **176** can maintain a correction if the lever reducer **102** is at one or more intermediate positions. The pawl **176** can be biased by the spring **172** acting on the lever arm **170**. The force of the spring **172** can facilitate maintaining the correction at one or more intermediate positions. The force of the spring **172** can facilitate maintaining the correction if force is no longer applied to the handle **120**.

[0106] In some methods, the lever reducer system **100** can be designed to lock the rod **112** to the first fastener **106** before engagement of the lever reducer **102**. In some methods, the lever reducer system **100** can be designed to lock the rod **112** to the first fastener **106** after removal of the lever reducer **102**. The first tower **104** can be aligned with the pivot point **200** during operation of the lever reducer **102**. The first fastener **106** and the first tower **104** can be aligned with the pivot point **200** during operation of the lever reducer **102**. Once the second fastener **110** and the second tower **108** are vertically lifted, the lever reducer **102** allows the threaded cap **198** to be inserted into the second tower **108** to secure the rod **112** to the second fastener **110**. In some methods, the rod **112** can be secured to the fasteners **106**, **110** before removing the lever reducer **102** from the towers **104**, **108**. In some methods, the rod **112** can be secured to the first fastener **106** before utilizing the lever reducer **102**. In some methods, the rod **112** can be secured to the second fastener **110** during use of the lever reducer **102**. In some methods, the rod **112** can be secured to the first fastener **106** after utilizing the lever reducer **102**.

[0107] While certain embodiments have been shown and described herein, it will be clear to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments described herein may be employed. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

Claims

1-22. (canceled)

23. A lever reducer system comprising: a first tower; a second tower; and a lever reducer comprising a base, a fulcrum, and a swivel, wherein the fulcrum is configured to be coupled to the first tower, wherein the swivel is configured to be coupled to the second tower, wherein as a proximal portion of the base proximal to the fulcrum is lowered, a distal portion of the base distal to the fulcrum raises to lift the second tower, wherein the swivel is configured to rotate relative to the distal portion of the base without translation.

24. The lever reducer system of claim 23, wherein the lever reducer further comprises a handle.

25. The lever reducer system of claim 24, wherein the handle is coupled to the proximal portion of the base.

26. The lever reducer system of claim 24, wherein the handle is angled relative to the base.

27. The lever reducer system of claim 23, wherein the fulcrum is configured to rotate relative to the

- base.
- 28.** The lever reducer system of claim 23, wherein the fulcrum is configured to translate relative to the base.
- 29.** The lever reducer system of claim 23, wherein an engagement feature is configured to allow rotational motion of the swivel in a first direction and prevent rotational motion of the swivel in a second direction, opposite the first direction.
- 30.** The lever reducer system of claim 29, wherein the engagement feature is configured to allow clockwise rotation of the swivel but prevent counterclockwise rotation of the swivel.
- 31.** The lever reducer system of claim 29, wherein the engagement feature comprises a ratchet.
- 32.** The lever reducer system of claim 23, wherein the swivel is configured to be coupled to an outer keyed surface of the second tower.
- 33.** The lever reducer system of claim 23, wherein a position of the swivel relative to the fulcrum is configured to allow for mostly vertical motion of the second tower.
- 34.** The lever reducer system of claim 23, wherein the swivel comprises one or more gears.
- 35.** The lever reducer system of claim 23, wherein a distance between a pivot point of the lever reducer and a pin of the swivel is constant during movement of the second tower.
- 36.** The lever reducer system of claim 23, wherein the fulcrum comprises a tapered end configured to be inserted into the first tower.
- 37.** A method of using a lever reducer system comprising: coupling a first tower to a first vertebra; coupling a second tower to a second vertebra; and coupling a lever reducer to the first tower and the second tower, wherein the lever reducer comprises a base, a fulcrum, and a swivel, wherein the fulcrum is coupled to the first tower, wherein the swivel is coupled to the second tower, lowering a proximal portion of the base proximal to the fulcrum such that a distal portion of the base distal to the fulcrum raises to lift the second tower, wherein the swivel rotates relative to the distal portion of the base without translation.
- 38.** The method of claim 37, wherein the lever reducer lifts a vertebra.
- 39.** The method of claim 37, wherein the lever reducer raises a fastener such that a rod is seated within a head of the fastener.
- 40.** The method of claim 37, wherein the fulcrum rotates relative to the base.
- 41.** The method of claim 37, wherein the fulcrum translates relative to the base.
- 42.** The method of claim 37, wherein the second tower is gripped by the swivel without obstructing a lumen of the second tower.
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